

CodeNova Business Website

Deployment & DevOps Document

Version: 1.0 **Project:** CodeNova Business Website

Defines the recommended source-control, environment, CI/CD, hosting, database migration, secrets, monitoring, backup and production deployment process for CodeNova.

1. Recommended Infrastructure

- GitHub for source control.
- Vercel for Next.js hosting.
- Supabase for PostgreSQL/Auth.
- Resend/Brevo for email.
- Google Analytics/Search Console for analytics/SEO.
- Optional external monitoring/error tracking.

2. Environments

- Development: local developer environment.
- Preview/Staging: generated from branches or pull requests.
- Production: live CodeNova domain.
- Use separate environment variables and credentials per environment.

3. Git Workflow

- main = production-ready.
- Feature branches for development.
- Pull requests for review.
- Run lint, type-check and tests before merge.
- Use protected production branch where possible.
- Tag production releases for traceability.

4. CI/CD Pipeline

- Install dependencies.
- Run linting.
- Run TypeScript/type checks.
- Run unit/integration tests.
- Run build.
- Run security/dependency checks.
- Deploy preview for pull requests.
- Deploy production after approved merge.

5. Deployment Steps

- Configure production domain.
- Configure hosting project.
- Configure environment variables.
- Connect Supabase project.
- Run database migrations.
- Deploy application.
- Verify HTTPS.
- Run smoke tests.
- Verify forms, admin login and analytics.
- Monitor after release.

6. Environment Variables

- NEXT_PUBLIC_SITE_URL.
- NEXT_PUBLIC_ANALYTICS_ID.
- SUPABASE_URL.
- SUPABASE_ANON_KEY.
- SUPABASE_SERVICE_ROLE_KEY — server only.
- DATABASE_URL if used.
- EMAIL_PROVIDER_API_KEY — server only.
- Calendar credentials — server only if enabled.

7. Database Migrations

- All schema changes must be versioned.
- Test migrations in preview/staging first.
- Back up production before risky migrations.
- Use forward migrations rather than manual production edits.
- Verify application compatibility before and after migration.

8. Secrets Management

- Never commit secrets to Git.
- Use Vercel/Supabase secret configuration or a dedicated secrets manager.
- Rotate compromised credentials immediately.
- Use separate development and production credentials.
- Restrict service-role keys to server-side code.

9. Domain & DNS

- Configure custom domain.
- Enable HTTPS/SSL.
- Configure www/root behavior consistently.

- Verify DNS records.
- Configure email domain authentication such as SPF/DKIM/DMARC if sending from a custom domain.
- Verify canonical site URL.

10. Monitoring

- Monitor uptime.
- Monitor server errors.
- Monitor failed API requests.
- Monitor database health.
- Monitor email failures.
- Review analytics and conversion events.
- Create alerts for significant production failures.

11. Backup & Recovery

- Use provider-supported automated database backups.
- Define recovery objectives appropriate to the business.
- Periodically test restoration.
- Keep important exported business data according to the retention policy.
- Document recovery steps.

12. Rollback Strategy

- Keep previous production deployment available for rollback.
- For code failures, redeploy the last known-good build.
- For database changes, use tested migration rollback or forward-fix procedures; do not improvise destructive changes during an incident.
- Document incident and recovery actions.

13. Production Smoke Tests

- Homepage loads.
- Services load.
- Desktop App service loads.
- Portfolio loads.
- Contact form works.
- Estimator works.
- Consultation flow works.
- WhatsApp CTA works.
- Admin login works.
- Database writes work.
- Analytics events fire.
- No console/server critical errors.

14. Maintenance

- Update dependencies regularly.
- Review security advisories.
- Review backups.
- Audit admin accounts.
- Review logs and monitoring.
- Review SEO/performance quarterly or after major releases.

15. Definition of Done

- Production deployment is reproducible.
- CI checks pass.
- Secrets are secure.
- Database is migrated.
- Domain and HTTPS are active.
- Monitoring is active.
- Smoke tests pass.
- Rollback path is documented.

Document Relationship

This document complements the CodeNova PRD, TRD, Security & Privacy Document and Frontend Requirements Document. Implementation should use the approved business information and actual production configuration rather than placeholders or unverified claims.